

MINOR PROJECT-Secure Multi-Tier Web Application Network Architecture on AWS

Project Objective

Design and implement a secure AWS network infrastructure using VPC, Public and Private Subnets, Security Groups, Route Tables, Internet Gateway, and Network ACLs (Firewalls).

Architecture

Internet -> Internet Gateway -> Public Route Table -> Public Subnet (Nginx Web Server) and Bastion Host -> Private Route Table -> Private Subnet (Database Server).

AWS Services Used

VPC, Public Subnet, Private Subnet, Internet Gateway, Route Tables, Security Groups, Network ACLs, EC2.

Implementation Steps

1. Create VPC (10.0.0.0/16)
2. Create Public Subnet (10.0.1.0/24)
3. Create Private Subnet (10.0.2.0/24)
4. Create and attach Internet Gateway
5. Configure Route Tables
6. Launch Web Server and Database Server

Security Groups

Web Server: Allow 22 from My IP, 80/443 from Internet.

Database: Allow 3306 from Web Server SG and 22 from Bastion SG.

Network ACLs



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Public Subnet: Allow 22,80,443.

Private Subnet: Allow 22,3306 and ephemeral ports.

Testing

Access Nginx webpage, verify private database is inaccessible from internet, connect via Bastion Host, test DB connectivity from Web Server.

Learning Outcomes

VPCs, Subnets, Route Tables, Internet Gateway, Security Groups, Network ACLs, Bastion Host, and multi-tier architecture.



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